

Teacher Guide — middle / module-02- panchabhuta

IKS Curriculum

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Overview

Module 2 — Panchabhūta (Five Elements)

Age band: Middle · **Grades:** 6–8 · **Duration:** 10 sessions × 45 min (≈ 2 weeks)

Learning objectives

By the end of this module, students will be able to:

1. **Recall** the five great elements (*pañca-mahābhūta*) — *prthvī*, *ap*, *tejas*, *vāyu*, *ākāśa* — with one observable example of each.
2. **Explain** the Sāṅkhya creation sequence (space → air → fire → water → earth) as a nested accumulation of qualities, using a diagram.
3. **Distinguish** between the *bhūta–tanmātra* mapping (each element associated with a perceptual quality: sound, touch, form, taste, smell) and the modern atomic/molecular framework.
4. **Apply** the five-element lens to classify everyday objects and observe imbalances in a local ecosystem (e.g. air pollution, water scarcity).
5. **Compare** the *pañcabhūta* framework with the modern states-of-matter framework, noting what each scheme captures that the other does not.

NEP 2020 alignment

- **§4.6** — Integration of Indian Knowledge Systems with mainstream science.
- **§4.23** — Experiential learning through sensory sorting and field observation.
- **§11.8** — Multidisciplinary linkage (science + Sanskrit terminology + ecology).
- **§4.27** — Local context: students observe their own surroundings as examples.

Prerequisites

- Familiarity with three states of matter (solid, liquid, gas) from middle-school science.
- Basic five senses vocabulary.
- No Sanskrit knowledge assumed; all terms introduced with IAST + meaning.

Key vocabulary

See `_shared/glossary.md` for full entries.

- *pañca-mahābhūta* — five great elements
- *pr̥thvī*, *ap*, *tejas*, *vāyu*, *ākāśa* — earth, water, fire, air, space
- *tanmātra* — subtle perceptual essence (introduced lightly at Middle band)
- *guṇa* — quality (used loosely as “property” at this band)

Materials checklist (whole module)

- 5 sample objects representing each element (rock, water jar, candle/lamp, balloon, empty box)
- Chart paper or pinboards for the element diagram
- Index cards (~60) for the sorting activity
- Smartphones or paper for the home-observation walk (Day 4 / homework)
- Project supplies depending on project format (poster paper, recyclables, audio recorder, etc.)

Day-by-day overview

Day	Title	Focus	Assessment
1	The Hook — Five Things Everywhere	Sensory introduction	—
2	Creation Sequence	How qualities accumulate as elements emerge	Exit ticket
3	Five Qualities and Five Senses	<i>tanmātra</i> mapping introduced	Formative quiz
4	Element Sort (activity)	Apply categories to real objects	Activity rubric
5	Mid-module Check	Consolidate + diagnostic	Formative quiz
6	Cross-disciplinary: Modern Science	Compare with states of matter & periodic table	Reflection
7	Group Work: Balance & Imbalance	Ecology and pollution as element-imbalance	Discussion
8	Project Session 1	Plan + first artefact	Project draft
9	Project Session 2	Build + rehearse	Project draft
10	Final Assessment + Showcase	Summative quiz + 3-min presentations	Summative + project

Citation sources used

- **vidya-karana RAG** queries cached in `.rag-cache/` — primary verses: SB 2.10.3, SB 3.5.36, SB 3.26.36, SB 3.26.44, SB 3.26.50 (Sāṅkhya cosmology and *tanmātra* enumeration).
- See `sources.md` for additional references.

Project brief (one-line teaser)

Each student (or pair) chooses one element and produces a poster, demo, short story, or 90-second video that shows **(a)** what the *pañcabhūta* tradition says about it, **(b)** one modern scientific parallel, and **(c)** one example of its imbalance in the student's own neighbourhood. Full brief in `assessments/project-brief.md`.

10-Day Lesson Plan

Module 2 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	The Hook — Five Things Everywhere	<code>days/day-01-hook.md</code>	Sensory station rotation
2	Creation Sequence	<code>days/day-02-creation-sequence.md</code>	Diagram-build, exit ticket
3	Five Qualities and Five Senses	<code>days/day-03-five-qualities.md</code>	Quality-to-sense matching; formative quiz
4	Element Sort	<code>days/day-04-element-sort.md</code>	<code>activities/activity-01-element-sort.md</code>
5	Mid-module Check	<code>days/day-05-mid-module-check.md</code>	<code>quizzes/formative.md</code> (Part A)
6	Cross-disciplinary: States of Matter	<code>days/day-06-cross-disciplinary.md</code>	Compare-contrast worksheet
7	Group Work: Balance & Imbalance	<code>days/day-07-group-discussion.md</code>	<code>activities/activity-03-element-walk.md</code> debrief
8	Project Session 1: Plan	<code>days/day-08-project-session-1.md</code>	Project draft submitted

Day	Session title	Lesson file	Activity / assessment
9	Project Session 2: Build	days/day-09-project-session-2.md	Project draft submitted
10	Final Assessment + Showcase	days/day-10-final-assessment.md	quizzes/summative.md + project

Pacing notes

- Days 1–3 are content-heavy. If your class needs a half-day buffer, split Day 3 across two sessions and roll Days 5–10 forward by one slot. Drop Day 6 if you must compress; it can be assigned as guided reading.
- Day 4’s element-sort activity benefits from advance setup — see the activity sheet for the 24-hour-ahead step (asking students to bring three “interesting” objects from home).
- Project work (Days 8–10) assumes you’ve decided pair vs. individual by Day 5.

Suggested running theme

Open each class with the same question on the board: **“Which element did you notice most yesterday — and why?”** Two students share for 30 seconds each. By Day 10 every student will have spoken at least twice. This builds the observation habit the module is designed to grow.

Formative vs. summative weight

- Formative quizzes (Days 3 and 5) are **not graded** — they are diagnostic. Use them to decide who needs scaffolding on Day 6.
- Summative quiz (Day 10) — see rubric in `assessments/rubric.md`. Combined with project, total 40 marks.

What students leave the module with

- A working five-element diagram in their notebook with their own examples.
- The five Sanskrit terms with meanings, in their own handwriting.
- One completed project artefact.
- The habit of noticing which elements dominate any given environment.

Teacher Notes

Teacher Notes — Module 2 Panchabhūta (Middle)

Posture for this module

Lead with what students can observe — the room, the air, the ground, light through the window, the empty space above their heads. The *pañcabhūta* framework is a way of organising those

observations, not a mythology and not a substitute for chemistry. Treat it as the kind of careful sorting that any culture’s natural philosophers do before the tools of modern measurement exist.

If a student asks “but isn’t this just wrong, because there are 118 elements?” — that’s an excellent question and it’s the entire point of Day 6. The two frameworks are answering different questions. The modern periodic table answers “what atoms exist?” The *pañcabhūta* framework answers “what categories of stuff and space do we live among?” Both are useful; neither obsoletes the other.

Common misconceptions to watch for

1. **“Fire is just hot gas — it isn’t an element.”** True in modern chemistry. In *pañcabhūta*, *tejas* refers to the *transformative principle* — heat, light, digestion (*jaṭharāgni*), metabolism. Not “fire” in the narrow chemical sense.
2. **“Ākāśa just means outer space.”** *Ākāśa* is closer to “the medium in which other things exist” — including the gap between two students sitting next to each other. The English word “space” works if students understand it broadly.
3. **“Earth = soil.”** *Prthvī* is the solid-state category as a whole — soil, rock, bone, wood, metal.
4. **“The Five Elements are the same as the Greek Four Elements (earth/water/fire/air).”** Mostly the same first four, but Empedocles’s scheme has no equivalent of *ākāśa*. The fifth element (“quintessence”) in Aristotle is heavenly aether, not space-in-general. Students at Senior band explore this contrast; at Middle band a one-sentence note is enough.
5. **Treating *tanmātra* as just “the five senses.”** They are the *objects* that the senses perceive — sound, touch, form, taste, smell — not the sense organs themselves.

On Sanskrit pronunciation

Drill these five terms aloud in week 1; students will refer to them throughout the module:

- *prthvī* — “PRITH-vee” (the *r* is a vowel: roll the *r* lightly)
- *ap* — “up” (rhymes with English “up”)
- *tejas* — “TAY-jus”
- *vāyu* — “VAH-yu”
- *ākāśa* — “AH-KAH-sha”

Differentiation hints

- **Tier up (fast finishers):** Ask them to find a sixth category that *pañcabhūta* doesn’t cover — and defend why it should or shouldn’t be added. (E.g. “electricity,” “information,” “darkness.”) This sets up the Senior-band comparison with modern field theory.
- **Tier down (scaffolding):** Reduce the five terms to three for the first lesson — focus on *prthvī*, *ap*, *vāyu* (the most concrete). Bring in *tejas* and *ākāśa* on Day 2.

Sensitivity notes

- Some students may come from families that read these texts devotionally. Honour that without privileging it. The framing is “this is one thing the tradition says; here is what science says; here is how to hold both.”
- Avoid implying that “ancient Indians discovered modern physics.” They didn’t — they did careful first-person observation and built a useful organising framework. That’s a real achievement; it doesn’t need inflation.

Project oversight (Days 8–10)

- Confirm groups/pairs by end of Day 5.
- On Day 8, every group should leave class with a one-paragraph plan and a chosen element.
- On Day 9, intervene if a group is still planning — they need to have a draft artefact by end of class.
- For Day 10 presentations, use the 3-minute timer strictly. The discipline of compression is part of the learning.

Assessment grading sequence

1. Mark Day 5 formative quiz immediately (during class while students do the sort activity warm-up the next day) so feedback lands within 24 hours.
2. Day 10 summative quiz can be marked over the weekend.
3. Project rubric should be discussed *with* students at the start of Day 8 — show them the rubric, let them ask questions.

If you only have 6 days, not 10

Compress as follows: - Day 1 (was Day 1) - Day 2 (was Day 2 + 3 condensed — drop the tanmātra-sense matching activity, keep the diagram) - Day 3 (was Day 4 element sort) - Day 4 (was Day 6 cross-disciplinary) - Day 5 (was Day 8 + 9 single-period project) - Day 6 (was Day 10 — summative quiz only, no presentation)

Skip Day 5 formative quiz and Day 7 group discussion in this compressed version.

Sources

Sources — Module 2 Panchabhūta (Middle)

All scriptural citations were retrieved via the **vidya-karana-kg** RAG and cached under `.rag-cache/`. The verses below are the primary anchors used across the lesson plan.

Primary scriptural sources (from RAG)

Citation	Used in	Content
SB 2.10.3	Day 1 hook, glossary	“bhūta-mātrendrya-dhiyām

Citation	Used in	Content
		<i>janma sarga udāhṛtaḥ ...</i> — defines <i>sarga</i> as the manifestation of the five gross elements (sky, air, fire, water, earth), their qualities, the senses, and the intellect.
SB 3.5.36	Day 2 creation sequence	The earth element emerges from water, carrying the quality of smell (<i>gandha-guṇām</i>), under the impulse of time.
SB 3.26.36	Day 3 qualities	“ <i>mṛdutvaṁ kaṭhīnatvaṁ ca śāityam uṣṇatvam eva ca ...</i> ” — softness, hardness, cold, heat are the attributes of touch (<i>sparśa</i>), the subtle element of air.
SB 3.26.44	Day 3 qualities	Taste (<i>rasa</i>) transforms into smell (<i>gandha</i>); water gives rise to earth, and smell is perceived by the nose.
SB 3.26.50 (paraphrase in cached chunk)	Day 2 creation, Day 3 qualities	Earth has five qualities, water four, fire three, air two, sky (space) only one — sound.

These are not exhaustive — the cache files under `.rag-cache/` contain additional supporting chunks from Canto 4, Canto 6, Canto 11, and Caitanya-caritāmṛta.

Modern science references (Day 6 cross-disciplinary)

These are *not* in either RAG; standard middle-school science textbooks suffice:

- **States of matter** — NCERT Class 9 Science, Chapter 1 (“Matter in Our Surroundings”). Solid / liquid / gas; plasma and Bose-Einstein condensate as further states.
- **Periodic table** — NCERT Class 9 Science, Chapter 5 (“The Fundamental Unit of Life”). For the distinction: in modern physics “element” = atom of a single proton-number; in *pañcabhūta* “element” = an observable category of matter/space.
- **Ecology and ecosystem balance** — NCERT Class 8 Science, Chapter 18 (“Pollution of Air and Water”).

Historical context references (Day 6 reading list, optional)

- Larson, G. J. (1979). *Classical Sāṃkhya: An Interpretation of its History and Meaning*. Motilal Banarsidass. — On the *tattva* enumeration in Sāṃkhya, of which the *pañcabhūta* are the last five.
- Plofker, K. (2009). *Mathematics in India*. Princeton University Press. — Sets the historical context of classical Indian intellectual frameworks.

(These are teacher references; not assigned to students at Middle band.)

Day 1 — The Hook: Five Things Everywhere

Module: Panchabhūta · **Band:** Middle (Grades 6–8) · **Time:** 45 min

Learning objective

By the end of this lesson, students will name the five great elements (*pañca-mahābhūta*) in Sanskrit and English and give one example of each from their own classroom.

Materials

- Five sample objects arranged at the front: a rock or brick (*prthvī*), a transparent jar of water (*ap*), an unlit candle or oil lamp (*tejas*), an inflated balloon (*vāyu*), an empty cardboard box with the flaps open (*ākāśa*).
- Board / chart paper.
- Five blank index cards per student.

Vocabulary introduced today

- *pañca-mahābhūta* (IAST: *pañca-mahā-bhūta*; lit. “five great elements”) — the framework that organises matter and space into earth, water, fire, air, and space.
- *prthvī*, *ap*, *tejas*, *vāyu*, *ākāśa* — the five elements themselves; see glossary.

Lesson flow

Warm-up (5 min)

- Teacher: “Look around this room without moving. What is everything you see *made of*? Don’t tell me the *name* — tell me what kind of stuff it is.”
- Take 4 quick answers. Don’t correct anyone. Write all answers on the board.

Core (20 min)

1. Reveal the five objects on the table one at a time. For each, ask: “*What is this an example of?*”
 - Rock → solid. “*In Sanskrit, this category is called prthvī — PRITH-vee — the earth-element.*” (Write on board.)
 - Water jar → liquid. “*Sanskrit: ap. (Some texts say jala.) The water-element.*”

- Candle → heat / light. “Sanskrit: tejas — TAY-jus — the fire-element. Not just a flame: anything hot, or that gives light, or transforms one thing into another.”
 - Balloon → air. “Sanskrit: vāyu — VAH-yu — the air-element. Anything that moves and flows like gas.”
 - Empty box → space. “And finally — the box* is made of cardboard, but what about what’s inside the box? The room you have? That’s* ākāśa — AH-KAH-sha — the space-element.”
2. **Key idea on the board:** > Around 1500 BCE, thinkers in the Indian subcontinent organised everything they could see into **five categories**. They called them the *pañca-mahābhūta* — the **five great elements**.
 3. Show the source: read aloud the SB 2.10.3 paraphrase from the cache file: > “... *bhūta* — the five gross elements (the sky, air, fire, water, earth) ...” > — Śrīmad-Bhāgavatam 2.10.3 (Read for adult ears, not for student memorisation. Just hear it once.)

Activity (15 min) — Five Things in This Room

- Hand out five index cards per student.
- Each card gets one element label at the top: *pr̥thvī*, *ap*, *tejas*, *vāyu*, *ākāśa* (with English in brackets).
- In 10 minutes, students walk around the room (or stay seated if space is tight) and write ONE example of that element they can see *right now*, in this room.
- *pr̥thvī* — the desk top, a textbook, the floor.
- *ap* — water in their water bottle, sweat on a friend’s brow, condensation on a window.
- *tejas* — sunlight through the window, the heat of a friend’s hand, the warmth of a tube light.
- *vāyu* — the breeze from a fan, their own breath.
- *ākāśa* — the space between two students, the gap above their heads.

Walk around. If a student writes “the chair” under *pr̥thvī*, that’s correct. If they write “a chair” under *ākāśa* and explain it’s the empty seat space, that’s also correct — and worth celebrating.

Wrap-up (5 min)

- Exit ticket (oral, going around the room): each student names ONE element and their own example. Anything that hasn’t been said yet earns a small cheer.
- Preview: “Tomorrow we’ll see how the tradition says these five elements *came to be* — one after another, in a specific order.”

Student-facing content

Today we learned: The Five Great Elements

The *pañca-mahābhūta* are the five categories ancient Indian thinkers used to organise everything around us:

Sanskrit (IAST)	English	One example from your room
<i>pr̥thvī</i>	Earth / solid stuff	_____
<i>ap</i>	Water / liquid stuff	_____
<i>tejas</i>	Fire / heat / light	_____
<i>vāyu</i>	Air / gas / movement	_____
<i>ākāśa</i>	Space / room	_____

Fill in the table with your own examples. Keep this in your notebook — we’ll add to it through the module.

Homework (if any)

- Bring **three interesting objects from home** to class tomorrow. They should be small enough to fit on your desk. (You’ll need them on Day 4.)
- Bonus: which element does each of your objects belong to?

Differentiation

- **One tier up:** Some objects belong to more than one element. Find an object in this room that fits *three* elements and explain why.
- **One tier down:** If five terms feel too many, focus on remembering three for now: *pr̥thvī*, *ap*, *vāyu* (earth, water, air). We’ll add the other two over the next two days.

Teacher notes

- Some students will immediately try to “win” by giving exotic examples. Gently steer them back: the win is *noticing*, not exotic-ness.
- If a student says “*tejas* doesn’t fit anything in this room because nothing is on fire,” ask them: is the tube light hot? Where is the warmth from? That’s *tejas*.
- Watch for confusion on *ākāśa*. Several students will insist “there’s nothing here” — say “yes, and that ‘nothing-here’ has a name, and the name is *ākāśa*.” It’s the medium other things sit in.

Citation(s) used in this lesson

- “SB 2.10.3” — vidya-karana RAG cache day-01-hook.json#chunks[0] (verse defining *sarga* and enumerating the five gross elements)

Day 2 — The Creation Sequence

Module: Panchabhūta · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will draw a diagram showing the five elements emerging in the Sāṅkhya creation sequence (space → air → fire → water → earth) and label what new quality each one adds.

Materials

- Board / large chart paper
- Coloured markers (5 colours, one per element)
- Each student needs: their notebook, a ruler

Vocabulary introduced today

- *guṇa* (IAST: guṇa; lit. “thread, quality, attribute”) — at this band, treat as “quality” or “property.”

Lesson flow

Warm-up (5 min)

- Question on the board (same daily prompt): “Which element did you notice most yesterday? Why?” — 2 students share for 30 seconds each.
- Quick recall: hold up each of the 5 objects from Day 1; students call out the Sanskrit name in chorus.

Core (20 min)

1. **Teacher prompt:** “If you had to *invent* the universe — start with nothing and make all matter — which element would you make *first*? Which *last*? Why?”
 - Take 3–4 student answers. Don’t judge them. Note: most will say earth first or water first.
2. **Reveal the Sāṅkhya answer (write on board, large):**

ākāśa → vāyu → tejas → ap → pṛthvī
(space) (air) (fire) (water) (earth)

“The tradition says the *opposite* of what most of us guess. Why might that be?”

3. **Explain with the quality-counting rule** (this is the key idea — write it boxed on the board):

Each element carries forward all the qualities of the elements before it, and adds one new quality of its own.

Build the table on the board, filling row by row:

Element	Qualities
ākāśa (space)	1 — sound
vāyu (air)	2 — sound, touch
tejas (fire)	3 — sound, touch, form
ap (water)	4 — sound, touch, form, taste
pṛthvī (earth)	5 — sound, touch, form, taste, smell

Source paraphrase: “Earth has five qualities, water has four, fire has three, air has two, and the sky has only one quality, sound.” — Śrīmad-Bhāgavatam 3.26.50

4. **Check:** “Why does it make sense that *earth* has all five qualities? Why does it make sense that *space* has only one?”
- Push toward: solid things are perceivable in *every* way — you can hear, touch, see, taste, smell soil. Empty space has no smell, taste, form, or touch — but you can hear sound in it.

Activity (15 min) — Draw your own diagram

- Students draw the five elements in their notebooks as a nested set of expanding “rings” or “stacked layers” — whichever visual makes sense to them.
- Each layer is labelled with its element AND its quality-count.
- Use coloured markers if available.
- At the bottom, each student writes a one-sentence answer: **“In my drawing, the reason X is the outer ring is ____.”**

Walk around. If a student draws *earth* as the smallest, “central” element, that’s also defensible — celebrate the reasoning, and gently distinguish: the SB sequence is about *order of appearance*, not size or importance.

Wrap-up (5 min)

- Exit ticket (written, 1 sentence, on a slip of paper turned in): *“If a friend asked you ‘why does space come first?’, what would you say?”*
- Preview: “Tomorrow we’ll meet the qualities themselves — sound, touch, form, taste, smell — and see how each one connects to one of your five senses.”

Student-facing content

The Five Elements emerge in a sequence

ākāśa (space) → *vāyu* (air) → *tejas* (fire) → *ap* (water) → *pṛthvī* (earth)

Each new element carries forward the qualities of the ones before it and adds one new quality of its own. So earth has all five, while space has just one — sound.

“mahīm gandha-guṇām ādhāt ...” — “The earth was endowed with the quality of smell, by the impulse of time.” — Śrīmad-Bhāgavatam 3.5.36

In your notebook, draw the five elements as nested layers, label each with its qualities, and write one sentence explaining why this order makes sense to you.

Homework

- Find ONE thing in your home that perfectly demonstrates the “all five qualities” of *pṛthvī*. You should be able to hear it, touch it, see it, taste it (or imagine the taste), and smell it. Bring a 2-sentence description.

Differentiation

- **One tier up:** Modern physics has electric and magnetic fields — would they fit into this sequence? If so, where? Defend your answer.

- **One tier down:** Just remember the order — space, air, fire, water, earth — and that earth has the most qualities. The “why” can come on Day 3.

Teacher notes

- The quality-counting rule is mathematically elegant and is the single best hook for students who like patterns. Linger here if your class is enjoying it.
- A student may ask: “But doesn’t fire have a smell? Doesn’t water have a colour?” — Good question. The SB framework is talking about the *defining* quality each element *adds*; we still perceive lower qualities in higher elements (e.g. water has all four — sound, touch, form, taste — but its *defining* contribution is taste/coolness).
- Sanskrit *guṇa* has a deeper meaning in Sāṅkhya philosophy (sattva/rajas/tamas). At Middle band, don’t open that door — say “we’ll meet a different meaning of *guṇa* in a later module.”

Citation(s) used in this lesson

- “SB 3.26.50” (paraphrase) — vidya-karana RAG cache day-02-creation.json#chunks[2] (Sāṅkhya quality-count rule)
- “SB 3.5.36” — vidya-karana RAG cache day-02-creation.json#chunks[1] (earth element acquiring smell as the last new quality)

Day 3 — Five Qualities and Five Senses

Module: Panchabhūta · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will match each *tanmātra* (subtle quality) to its element and to the sense organ that perceives it, and explain why this mapping is symmetric.

Materials

- Board with a blank 3-column table ready
- Optional: small samples for the senses station — a bell (sound), a piece of cotton (touch), a coloured leaf (form), a piece of jaggery or salt crystal (taste), a sprig of mint or a slice of orange peel (smell)
- Notebook, pen

Vocabulary introduced today

- *tanmātra* (IAST: tanmātra; lit. “that-much, subtle essence”) — the perceptual quality that “belongs” to each element. Five of them: *śabda* (sound), *sparsa* (touch), *rūpa* (form), *rasa* (taste), *gandha* (smell).

Lesson flow

Warm-up (5 min)

- Daily prompt: “Which element did you notice most yesterday?” — 2 students share.
- Hold up the bell. Ring it once. “What did you just perceive?” (Sound.)
- “Now — through what element did the sound reach you?” (Air, and ultimately space.)

Core (20 min)

1. **Build the master table on the board** (one row at a time, asking students to predict the senses):

Element	Tanmātra (quality)	Sense organ	Test it:
<i>ākāśa</i> (space)	<i>śabda</i> (sound)	ear	Ring the bell
<i>vāyu</i> (air)	<i>sparsā</i> (touch)	skin	Feel a breeze on your hand
<i>tejas</i> (fire)	<i>rūpa</i> (form / colour)	eye	Look at a flame’s colour
<i>ap</i> (water)	<i>rasa</i> (taste)	tongue	Taste sweetness / saltiness
<i>prthvī</i> (earth)	<i>gandha</i> (smell)	nose	Smell soil, or jaggery, or peel

2. **Connect to yesterday’s quality-count rule:** the *tanmātra* added at each element is the new quality that earth carries all five of. So earth — solid stuff — is the one you can perceive in every way.

3. **Source check.** Read (don’t memorise):

“*mṛdutvaṃ kaṭhinatvaṃ ca śaityam uṣṇatvaṃ eva ca / etat sparsāsya sparsātvaṃ tanmātratvaṃ nabhasvataḥ*” “Softness, hardness, cold and heat — these are the qualities of touch, which is the subtle attribute (*tanmātra*) of air (*nabhasvat*).” — Śrīmad-Bhāgavatam 3.26.36

“Notice what the *tanmātra* of touch is — not just ‘softness’ but also ‘cold and heat.’ The tradition is sorting *physical sensations* into the air-element bucket. Does that surprise you?”

4. **Modern parallel (one sentence each, no deep dive yet — that’s Day 6):**

- Modern physiology: sound = pressure waves transmitted through a medium → ear drum. Touch = mechanoreceptors in skin. Form = light reflected off objects → retina. Taste = chemical receptors on tongue. Smell = airborne molecules → olfactory receptors.

- Both systems are sorting our perception. The Indian system asks “what kind of *stuff* does this perception live in?” Modern science asks “what physical process produces this perception?”

Activity (15 min) — Sense Station Rotation

- 5 stations around the room (or 5 desks, one per element). Each has:
 - A label (Sanskrit + English + tanmātra + sense organ)
 - A small sample to perceive (see materials list)
 - A worksheet question
- Students work in pairs. 2 minutes per station. They rotate clockwise.
- At each station, they answer in one sentence: “*What did I perceive? What other element could I perceive at the same time?*” (E.g. at the bell station: heard the sound — also felt the vibration if they touched the bell.)
- **End-of-rotation discussion (3 min):** Did any sample give you *more than one* perception? Which? What does that tell us about how the elements overlap in the real world?

Wrap-up (5 min)

- **Formative quiz on a slip of paper (5 questions, 3 min)** — see quizzes/formative.md Part A.
- Collect for grading; the next day’s class includes the result.

Student-facing content

Each element has a quality. Each quality is perceived by a sense.

Element	Quality (<i>tanmātra</i>)	Perceived by
<i>ākāśa</i> (space)	<i>śabda</i> (sound)	ear
<i>vāyu</i> (air)	<i>sparsa</i> (touch)	skin
<i>tejas</i> (fire)	<i>rūpa</i> (form, colour)	eye
<i>ap</i> (water)	<i>rasa</i> (taste)	tongue
<i>prthvī</i> (earth)	<i>gandha</i> (smell)	nose

Earth has all five qualities. That’s why you can hear soil shift, feel its texture, see its colour, taste its mineral content (you’ve all chewed sand in childhood — admit it), and smell it after rain.

“Softness, hardness, cold and heat are the qualities of touch, the subtle attribute of air.”
— Śrīmad-Bhāgavatam 3.26.36

Homework

- For each of your three home-objects (which you’ll bring on Day 4), figure out the *dominant* element and the *dominant tanmātra*. Write one line per object.

Differentiation

- **One tier up:** What about ultrasonic sound (above 20,000 Hz) — humans can't hear it, but bats can. Does it still belong to *ākāśa*? What about magnetic fields, which no human sense can detect directly?
- **One tier down:** Focus on memorising the table. The “why does this match the order of creation?” question can come later.

Teacher notes

- Some students will challenge “fire = form/colour.” Walk them through: the *defining* perception of fire is what we *see* — light, colour. We feel heat from fire, but that's borrowed from *vāyu*'s tanmātra of touch (heat). The mapping is about the *new* quality each element introduces.
- Watch for the smell-station: students will spend most of their time there. That's fine. Give them 30 extra seconds and pull them along.
- If using jaggery / sweets — check for diabetic students. Substitute with salt or a non-edible smell-only sample.

Citation(s) used in this lesson

- “SB 3.26.36” — vidya-karana RAG cache day-03-qualities.json#chunks[3] (tanmātra of touch, qualities of air)
- “SB 3.26.44” — vidya-karana RAG cache day-03-qualities.json#chunks[0] (rasa transforms into gandha; smell perceived by nose)

Day 4 — Element Sort (Hands-on Activity)

Module: Panchabhūta · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will classify at least 12 everyday objects by their dominant element and defend a classification where more than one element is present.

Materials

- Students bring their 3 home-objects from yesterday's homework.
- ~5 mystery objects supplied by the teacher (suggested: a bicycle pump, a glass of milk, a magnifying glass, a small mirror, a sponge).
- 5 sorting trays / boxes / labelled areas on the floor — one per element.
- Worksheet (one per pair) — printable from [activities/activity-01-element-sort.md](#).

Vocabulary recap

No new terms today. Review *prthvī*, *ap*, *tejas*, *vāyu*, *ākāśa* and the five tanmātras.

Lesson flow

Warm-up (5 min)

- Daily prompt: “Which element did you notice most yesterday?” — 2 students.
- Brief: today is a sorting day. We’ll see how messy the world really is when you try to put it in five boxes.

Core (10 min) — Demonstrate with one tricky object

- Hold up a glass of milk. “Where does this go?”
- Take 3 student answers.
- Then walk through it together:
 - It’s a *liquid* → *ap* (water-element).
 - But milk has *fat* (solid suspended particles) → some *pr̥thvī*.
 - It has *taste* (*rasa*) → *tanmātra* of water, confirms.
 - It has *smell* (*gandha*) → some earth-quality too.
 - You can see it (*rūpa*) → some fire-quality (form/colour).
- Conclusion: **most real objects are mixtures**. The exercise is to identify the *dominant* element while honestly naming the others present.

Activity (25 min) — Element Sort

Follow the full activity sheet ([activities/activity-01-element-sort.md](#)). Brief outline:

1. Students work in **pairs**.
2. Each pair places their 3 home-objects + 2 teacher-supplied objects on the right element tray. (5 objects per pair.)
3. For each object, they fill the worksheet table:
 - Object name
 - Dominant element
 - Other elements present (1–4 more)
 - Strongest *tanmātra*
4. **Gallery walk (5 min within the 25):** pairs visit two other pairs’ trays and put a sticky note where they disagree, explaining briefly.

Wrap-up (5 min)

- Discussion question, open to the room: “Which object was the hardest to classify, and why?”
- Preview Day 5: a mid-module check + an extended reflection.

Student-facing content

(The activity sheet is the student-facing content — see [activities/activity-01-element-sort.md](#).)

Homework

- No new homework. (Spend the freed time finalising the project pair or topic — partners and projects are confirmed by Day 5.)

Differentiation

- **One tier up:** After sorting, pick the **single most “mixed”** object and write a paragraph explaining the proportions you’d assign each of the five elements within it.
- **One tier down:** If five categories feel hard, sort first into THREE — *pṛthvī*, *ap*, *vāyu* — then come back and consider where *tejas* or *ākāśa* should be added.

Teacher notes

- This activity often surfaces interesting disagreements between students who classify the same object differently. **Don’t resolve the disagreements quickly** — let them argue it out. The reasoning is more important than the answer.
- A common dispute: where does a “balloon filled with air” go? *Vāyu* because the *content* is air? Or *pṛthvī* because the *balloon material* is rubber (a solid)? Both are defensible — push students to articulate which one is “dominant.”
- Look for students who are insisting on a single rigid answer. They are missing the lesson — gently steer toward “and what’s the next-strongest element here?”
- Photograph any especially good or disputed sorting choices; reuse on Day 7 group discussion.

Citation(s) used in this lesson

- No new citations introduced. Review citations from Day 1–3.

Day 5 — Mid-Module Check

Module: Panchabhūta · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will demonstrate working recall of all five elements, their qualities, and the creation sequence, and will commit to a project topic and partner.

Materials

- Printed formative quiz (5 questions, see `quizzes/formative.md` Part B) — one per student.
- Sample project briefs (see `assessments/project-brief.md`) — one per pair.
- Sticky notes (3 per student).

Lesson flow

Warm-up (5 min)

- Hand back yesterday's formative slip (the 5 quick questions from Day 3). Don't review answers — just hand them back.
- Daily prompt: “Which element did you notice most over the weekend?” — 2 students.

Core (20 min) — Formative quiz + class review

1. **Formative quiz (10 min)** — quizzes/formative.md Part B. Closed notebook. 5 questions: MCQ on the order of emergence, fill-in on the quality-table, one short-answer “name an object dominated by *ākāśa*.”
2. **Review together (10 min)** — go through each question. Don't grade now; grade overnight. The goal here is exposure to common errors. Ask: “who would have answered (b)? Why?” Invite the wrong reasoning, surface the misconception, correct it.

Activity (15 min) — Project commitment

- Hand out the project brief. Read aloud the **one-paragraph project description** and the **list of project format options** (poster, demo, short story, 90-second video, infographic, songbook).
- Pairs / individuals choose:
 - Element they'll focus on
 - Format
 - One-sentence “what we'll show”
- Each pair writes their three choices on a sticky note and posts on the board under the element label.
- Teacher does a 30-second sanity check on each — if a pair has chosen something impractical for the time available, redirect now.

Wrap-up (5 min)

- The board is a snapshot of the class's project distribution. Note out loud: “We have 4 groups on water, only 1 on space — interesting. Group on space, you'll have a unique presentation. Water groups, find a way to make yours distinct.”
- Preview Day 6: “Tomorrow we leave Sanskrit for a day and look at the modern science framework — and ask which one is ‘right.’”

Student-facing content

Mid-module check — you'll take a short quiz today and choose your project.

Project options for your final assignment (Days 8–10):

Format	Best for elements ...
Poster (A2)	Any
Live demo (2 min)	<i>tejas</i> (fire), <i>vāyu</i> (air)
Short story (1 page)	Any — especially <i>ākāśa</i>

Format	Best for elements ...
90-second video	Any
Infographic	<i>pr̥thvī</i> , <i>ap</i> (data-heavy)
Original song / poem	<i>vāyu</i> , <i>ākāśa</i>

You'll work in pairs or alone. Decide today.

Homework

- Submit project plan in writing by next morning. Format: 1 paragraph stating element, format, and what audience will see/hear/feel.
- Begin gathering project materials.

Differentiation

- **One tier up:** A pair tackling space (*ākāśa*) — challenge: how do you make a poster about emptiness? Force the constraint and find a creative answer.
- **One tier down:** Allow individual work with teacher check-ins every 10 minutes during Days 8–9.

Teacher notes

- Grade the formative quiz tonight and walk in tomorrow knowing who is shaky on the creation sequence vs. who is shaky on the tanmātra mapping. The two error patterns need different reteaching.
- If too many pairs cluster on one element (e.g. 6 pairs on fire), gently broker — “groups 4 and 5, would you consider switching to space or air? You’ll have less competition for attention on presentation day.”

Citation(s) used in this lesson

- No new citations introduced.

Day 6 — Cross-Disciplinary: Pañcabhūta and Modern Science

Module: Panchabhūta • **Band:** Middle • **Time:** 45 min

Learning objective

By the end of this lesson, students will compare the *pañcabhūta* framework with the modern states-of-matter / periodic-table framework, identifying what each captures that the other does not.

Materials

- A copy of the standard periodic table on the wall (any classroom poster works).
- A printed *pañcabhūta* table (the one built on Day 3).

- A two-column comparison worksheet (blank, students fill in).

Vocabulary recap

No new Sanskrit. Today is the day to use English freely and compare frameworks.

Lesson flow

Warm-up (5 min)

- Daily prompt — 2 students.
- Provocation: “If I told you ‘fire is not an element,’ would I be right or wrong?”
- Take both sides. Don’t resolve. “Today we’ll see how both answers can be correct, because they’re answering different questions.”

Core (25 min)

1. **Two frameworks, two questions** — put on the board:

Framework	What it sorts	The question it answers
<i>Pañcabhūta</i> (c. 1500 BCE onwards)	Everyday matter and space	“What categories of stuff do I live among?”
Modern periodic table (1869 onwards)	Atoms	“What kinds of atoms exist?”

2. **Side-by-side comparison.** Build this table together on the board (students copy):

Property	<i>Pañcabhūta</i>	Modern science
Number of categories	5	118 elements; ~4 states of matter
Includes empty space?	Yes (<i>ākāśa</i>)	No (space is the <i>absence</i> of matter)
Distinguishes heat/light?	Yes (<i>tejas</i> is its own element)	No (energy is not an element; fire is hot gas)
Distinguishes liquid from solid?	Yes (<i>ap</i> vs <i>prthvī</i>)	Yes (states of matter)
Distinguishes hydrogen from helium?	No (both are <i>vāyu</i>)	Yes (different atoms)
Method	Sensory observation	Experiment + measurement
Useful for	Sorting your everyday environment, ecology, Ayurveda	Chemistry, materials science, physics

3. **Important framing (write large on the board):**

It is not a competition. A modern chemist still uses the words *solid, liquid, gas*. Modern Indian classical music still teaches *śabda*. Both frameworks survive because both are useful.

4. One worked example: a candle flame.

- *Pañcabhūta* lens: dominantly *tejas* (heat + light + transformation). Touching air (*vāyu*) which feeds it. Made possible by *ākāśa* around it.
- Modern chemistry lens: combustion reaction — hydrocarbon (wax) + O₂ → CO₂ + H₂O + light + heat. The “flame” is incandescent gas (mostly soot particles glowing).
- Same object, two complete descriptions. Neither one wrong.

Activity (10 min) — Compare-Contrast Worksheet

Students pick **any 3 objects** from the Day 4 sort. For each, they fill in a 2-column row: - **Pañcabhūta description** (which element dominates, which tanmātra, which others present) - **Modern science description** (state of matter, dominant chemical components if known)

They then write one sentence: “*For this object, the framework I’d choose to describe it depends on ...*”

Wrap-up (5 min)

- Cold-call 3 students to share their “I’d choose framework X because ...” sentence.
- Note out loud: there’s no universal right answer. The framework depends on the question.
- Preview Day 7: “Tomorrow we’ll look at what happens when one of the five elements is *out of balance* in a real ecosystem — and what that has to do with us.”

Student-facing content

Two frameworks for the same world

The *pañcabhūta* tradition organises matter and space into 5 categories using sensory observation. Modern science organises matter into 118 elements (and 4+ states of matter) using experiments and measurement. They are answering different questions.

Question	Framework that fits
“What’s the dominant <i>feel</i> of this object?”	Pañcabhūta
“What atoms make up this object?”	Periodic table
“Why does soil smell after rain?”	Both, in different ways

Both frameworks are still in use today. Neither one is “wrong.”

Homework

- Choose **one** of the modules from your science textbook chapter on Matter (states of matter, or the periodic table). Write a 1-paragraph note: *what would the pañcabhūta framework get right about this topic, and what would it miss?*

Differentiation

- **One tier up:** Read about *plasma* as a fourth state of matter. Where would *plasma* fit in *pañcabhūta*? Defend your placement.
- **One tier down:** Just remember the framing: *both frameworks survive because both are useful*. The comparisons can be filled in over the next few days.

Teacher notes

- This is the day skeptical students will engage most. Welcome it. Do not retreat into “ancient Indians were right too” — that defeats the lesson. The honest stance is *both frameworks are real intellectual achievements with different scopes*.
- If a student says “the *pañcabhūta* framework is incomplete because it can’t predict chemistry” — they’re correct. Acknowledge that. Then ask: “Can the periodic table predict whether the air in this room feels stuffy or fresh? Both frameworks have limits.”
- For NEP §4.6 compliance, the framing is integration, not replacement. The line to hold: students are learning *two* useful frameworks, not one true one.

Citation(s) used in this lesson

- No new RAG citations introduced.
- Modern science references: NCERT Class 9 Science, Chapters 1 (“Matter in Our Surroundings”) and 5 — see `sources.md`.

Day 7 — Group Work: Balance and Imbalance

Module: Panchabhūta · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will identify a real local example of one *pañcabhūta* being “out of balance” in their environment and propose one observable consequence and one mitigation action.

Materials

- 5 large pieces of chart paper, one labelled with each element.
- Sticky notes (~30) per group.
- Markers.
- Optional: 3–5 short newspaper clippings about local environmental issues (air quality, water shortage, waste/landfill, heatwave, deforestation).

Vocabulary

- No new Sanskrit. Introduce the word **imbalance** as the operating concept of the day.

Lesson flow

Warm-up (5 min)

- Daily prompt — 2 students.
- “All week we’ve talked about the elements as if they sit nicely in their boxes. Today we’ll see what happens when one of them takes over — or disappears.”

Core (15 min) — From individual elements to imbalanced systems

1. **Open question, take 3 answers:** “Can you think of a place where one of the five elements is too much, or too little?”
 - Example answers students typically give: too much smoke (*tejas* or *vāyu* — pollution), no water (*ap* — drought), too much plastic waste (*pr̥thvī* — landfill), too noisy a road (*ākāśa* + *śabda* — sound pollution), too hot a city (*tejas* — heat island).
2. **Frame the principle (write on board):**

A healthy environment has all five elements in balance. When one is overloaded or depleted, the others suffer.
3. **Mini case study together — air pollution in Indian cities:**
 - Too much *tejas* in the form of vehicle exhaust → *vāyu* becomes carrying polluted matter → *ap* (rain) becomes acidic → *pr̥thvī* (soil) absorbs the acid → plants struggle → less oxygen → *vāyu* worsens further.
 - Notice: an imbalance in one element cascades.

Activity (20 min) — Five-Element Imbalance Map

In groups of 4, each group tackles ONE element. On their large chart paper they fill four quadrants:

- **What this element does for us** (positive contributions)
- **What an excess looks like in our area** (one specific local example)
- **What a depletion looks like in our area** (one specific local example)
- **One small thing students our age can actually do** (concrete, doable this week)

Each group sticks their chart on the wall. Gallery walk: students rotate, read all five charts, and place a sticky note on the action they’d most commit to.

Wrap-up (5 min)

- Each group reads aloud their chosen action (just the one line).
- Note: “Notice how no single action solves everything — but every chart has at least one thing that’s doable.”
- Preview Day 8: project work begins. Bring all your materials tomorrow.

Student-facing content

Balance and imbalance

A healthy environment has all five elements in balance. When one takes over or disappears, the others suffer.

Examples of imbalance you might recognise:

Imbalance	What you observe
Too much <i>tejas</i> (heat)	Urban heat-island; summer power cuts
Too little <i>ap</i> (water)	Drought; failed crops; long water-queue at the tap
Polluted <i>vāyu</i> (air)	Smog; difficulty breathing on bad-air days
Polluted <i>pṛthvī</i> (earth)	Plastic waste; barren soil; landfill near homes
Noisy <i>ākāśa</i>	Sound pollution; sleep disturbance; bad concentration

The Sanskrit concept *Vasudhaiva Kuṭumbakam* — “the earth is one family” — is the ecological framing for this. We’ll meet it again in Module 13.

Homework

- Bring **all** project materials to tomorrow’s session. The work begins in earnest.
- For your project, consider: is there a small element of “imbalance” worth showing alongside the element’s main story?

Differentiation

- **One tier up:** Pick TWO elements and explain how an imbalance in one causes an imbalance in the other. Be specific.
- **One tier down:** Focus on choosing ONE concrete action you can commit to this week (carry a water bottle to reduce plastic, switch off lights when leaving a room, etc.). The systems-thinking can come later.

Teacher notes

- This lesson connects to NEP §11.8 (multidisciplinary linkage) and is a natural lead-in to Module 13 (Indic Ecology) when the time comes.
- Some students may feel discouraged by the scale of the problems. Don’t soften the reality, but anchor every imbalance to a doable response. The point is agency, not despair.
- If your school is in a location with a *specific* well-known environmental issue (e.g. a polluted river, a heavily quarried hillside), centre that as the case study and skip the air-pollution example.

Citation(s) used in this lesson

- *Vasudhaiva Kuṭumbakam* is foreshadowed but not cited. Save full sourcing for Module 13.
- No new RAG citations.

Day 8 — Project Session 1: Plan and Begin

Module: Panchabhūta · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, every pair / individual will have a written project plan and at least one tangible artefact (draft poster outline, demo prop, story opening paragraph, etc.) for their final project.

Materials

- Project materials brought by each pair (as per their plan)
- Spare poster paper, markers, scissors, glue
- A printed copy of the rubric (`assessments/rubric.md`) per pair
- Stopwatch / classroom timer

Lesson flow

Warm-up (5 min)

- Daily prompt — 2 students.
- Hand out the rubric. Walk through it in 60 seconds: *“You’ll be graded on accuracy, use of sources, originality, communication, collaboration, and presentation. Highest mark in each row is 4.”*

Core (35 min) — Workshop

Structure: Three 10-minute sprints with a 1-minute “stand and stretch” between each.

- **Sprint 1 (10 min) — Plan completion.** Every pair finalises:
 - Element + project title
 - The single sentence the audience should walk away thinking
 - The structure (3 parts max: e.g. “what tradition says / one science parallel / one local imbalance”)
 - The division of labour (who does what)
 - The source(s) they will cite (must include one RAG citation from Day 1–3 caches, OR a citation from `sources.md`)
- **Sprint 2 (10 min) — First artefact.** Pairs start building. Teacher does rapid-rotation: 90 seconds per pair, asking one focused question.
- **Sprint 3 (10 min) — Build / iterate.** Continue.

Teacher’s role during sprints: - 30-second visit to each pair in Sprint 2. - Watch for pairs who are *still planning* during Sprint 2. Pull them aside, force a “start drawing / start writing / start building NOW” intervention. - For projects on *ākāśa* (often the trickiest), check that they’ve found a concrete way to show emptiness — silence, white-space, a pause, a gap.

Wrap-up (5 min)

- 30-second share-out from any 4 pairs (you call them) — they hold up their progress.
- Submit project plan paragraph at the door on the way out.
- Preview Day 9: refine and rehearse.

Student-facing content

Project workshop — Session 1

Today you'll plan and start your project. By end of class, your pair should have:

1. A written plan (the paragraph you turn in today).
2. Your first concrete artefact in progress — even if rough.

Plan template:

Element: _____

Title: _____

Audience walks away thinking: _____

Structure (3 parts):

1. _____
2. _____
3. _____

Division of labour (Partner A): _____

Division of labour (Partner B): _____

Sources we'll cite:

- _____ (from class cache or sources.md)

Homework

- Whatever your pair didn't finish today — finish at home tonight. Tomorrow is for *refining*, not for *starting*.

Differentiation

- **One tier up:** Stretch goal — add an interactive element (a question for the audience, a small experiment they can join). Adds depth.
- **One tier down:** Allow choosing a *simpler* format mid-stream if the original plan is collapsing. A clear poster beats a half-built video.

Teacher notes

- The pacing of three sprints is deliberate — most students will resist starting to build. The 10-minute timer forces them past planning paralysis.
- If a pair seems stuck, ask: “*If you had to present right now, in 30 seconds, what would you say?*” That often unlocks the structure.

Citation(s) used in this lesson

- The pair's chosen source(s) — verified during Sprint 1.

Day 9 — Project Session 2: Refine and Rehearse

Module: Panchabhūta · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, every pair will have a presentation-ready project artefact and will have rehearsed their 3-minute presentation at least once with timed feedback.

Materials

- Pairs bring all in-progress materials.
- Stopwatch.
- Printed feedback form: 1 strength, 1 fix, in 30 seconds.

Lesson flow

Warm-up (5 min)

- Daily prompt — 2 students.
- Brief: “Today is **refine + rehearse**. By end of class, you’ll have presented to one other pair and gotten feedback.”

Core (30 min)

Structure: Two 15-minute halves.

- **Half 1 (15 min) — Refine.**
 - Continue building / drafting.
 - Teacher does rapid rotation: 60 seconds per pair, focused question: “*Show me your strongest 30 seconds.*”
 - Look for projects that:
 - Are missing the citation — flag immediately.
 - Are missing the modern-science parallel — required for accuracy mark.
 - Are missing the local imbalance — required for the “originality / connection” mark.
 - Pairs that look ready early start to rehearse.
- **Half 2 (15 min) — Rehearse with a buddy pair.**
 - Match each pair to one other pair. (Vary partners — match a strong pair with one that’s behind, etc.)
 - Pair A presents to Pair B for 3 minutes (use the timer).
 - Pair B gives feedback for 60 seconds — one strength, one fix.
 - Swap. (Total: 8 min per rotation, $\times 2 = 16$ min — runs slightly over, that’s OK.)

Activity in parallel

- Teacher with the timer; intervene if a pair tries to rehearse for more than 3 min.

- Listen for projects that are *all summary, no specifics*. Intervene: “Give me one concrete example or detail — what does the audience *see*?”

Wrap-up (5 min)

- One-sentence share: “*What’s the one thing I’m changing before tomorrow?*” — every pair contributes.
- Tomorrow: full presentations + summative quiz.

Student-facing content

Project workshop — Session 2: Refine + Rehearse

Today’s checklist:

- ☐ My artefact is presentation-ready (no half-finished sections).
- ☐ I’ve cited at least one source (verse, sūtra, or PDF).
- ☐ I’ve named at least one modern-science parallel.
- ☐ I’ve named one local example of imbalance.
- ☐ I’ve timed my presentation — it fits in 3 minutes.
- ☐ I know what my partner says and what I say.
- ☐ I’ve heard feedback from one other pair and decided what to change.

Tomorrow you present.

Homework

- Final touches at home tonight. Bring everything tomorrow.

Differentiation

- **One tier up:** Add a Q&A-ready note: anticipate one tricky question from the audience and prepare a 30-second answer.
- **One tier down:** Allow reading from a written script for the presentation. The goal is communication, not memorisation.

Teacher notes

- The “buddy pair” rehearsal often surfaces problems the teacher missed. Encourage students to be *kind and specific* in their feedback — “the part about water was clear, the part about pollution was rushed” beats “good job.”
- Pairs who consistently can’t fit in 3 minutes are usually trying to say too much. Help them cut.

Citation(s) used in this lesson

- Verified per pair during refine half.

Day 10 — Final Assessment + Project Showcase

Module: Panchabhūta · **Band:** Middle · **Time:** 45 min (potentially extend to 60 if presentations need it)

Learning objective

By the end of this lesson, students will demonstrate mastery of the module’s learning objectives through a summative quiz and a 3-minute project presentation.

Materials

- Printed summative quiz (`quizzes/summative.md`) — one per student.
- Stopwatch.
- Rubric copy (printed) for each pair, with teacher marking column.
- A “reflection slip” — one per student.

Lesson flow

Warm-up (3 min)

- No daily prompt today.
- Brief: “Two parts today — a quiz first, then your presentations. Quiz is 20 minutes, individual. Presentations after.”

Part 1: Summative Quiz (20 min)

- Distribute `quizzes/summative.md`. Closed-book. No discussion.
- Quiz covers: element names (Sanskrit + English), creation sequence, quality-count rule, tanmātra-to-sense mapping, one comparison question on pañcabhūta vs. modern science, one open-response on element imbalance.
- Collect quizzes at 20-min mark.

Part 2: Project Presentations (20 min)

- 3 min presentation + 1 min Q&A per pair.
- With ~8–10 pairs in a typical class, you’ll fit 5 pairs in the remaining 20 min. **Reserve the next class period for the remaining presentations.** (Note this honestly — Day 10 alone is rarely enough for 10+ pairs.)
- During each presentation:
 - One teacher marks the rubric in real-time.
 - One audience member is assigned to ask one specific question.
 - Audience completes a “1 thing I learned, 1 question I have” slip per presentation.

Wrap-up (2 min)

- Thank the class.
- Closing question (oral, voluntary): “*What’s one thing you’ll keep noticing after this module ends?*”

- Reflection slip due: students answer the three reflection questions from `assessments/project-brief.md`.

Student-facing content

Today: quiz + presentations

- 20 minutes for the summative quiz — closed book.
- 3-minute presentation per pair, then 1 minute of audience questions.

When you're in the audience, fill out one "I learned / I wonder" slip per presentation. These go to the presenters and help them know what landed.

Reflection (due at the end of class):

1. What did you set out to learn or make, and what did you actually end up with?
2. What is the most surprising thing you discovered while making this?
3. If you had three more days, what would you do differently?

Homework

- None. The module is complete. (Begin the next module's prerequisite reading if your sequence flows directly into another.)

Differentiation

- **One tier up:** A student who finishes the quiz early can write a 1-paragraph "what would Module 2 look like for a class of Grade 11 students?" — useful input for the Senior-band pack.
- **One tier down:** Allow the quiz to be open-notebook for a student with documented learning needs — your call.

Teacher notes

- Block out 30 minutes after class for marking the quiz while presentations are fresh.
- If presentations spill into Day 11, that's fine — protect that time. Don't compress presentations to fit; you lose the assessment validity.
- Mark the rubric in real-time, but write detailed comments only after each presentation finishes — the comment is what students keep.

Citation(s) used in this lesson

- All citations verified across the module.

Module 2 (Middle) — Formative Quizzes

Two parts: Part A on Day 3 (after qualities lesson), Part B on Day 5 (mid-module). Both are diagnostic — not graded. Hand back within 24 hours.

Part A — Day 3 (3 min, 5 questions)

Name: _____ Class: _____

1. Write the Sanskrit name for the **water** element.

2. Which sense organ perceives *rūpa* (form / colour)?

- a) ear
- b) skin
- c) eye
- d) tongue

3. Fill in the blank: “The *tanmātra* of *vāyu* is _____.”

4. True or false: *Ākāśa* has only one quality, which is sound. (T / F)

5. Name ONE object in this room dominated by *tejas*.

Part A — answer key

Q	Answer	Notes
1	<i>ap</i> (or <i>jala</i>)	Either is acceptable; <i>ap</i> preferred.
2	c) eye	—
3	<i>sparsā</i> (touch)	Spelling tolerated if recognisable.
4	T	—
5	tube light / sunlight through window / heater / candle / etc.	Any source of heat or light.

Part B — Day 5 (10 min, 5 questions, mixed format)

Name: _____ Class: _____

1. (2 marks) Write the five elements in the order they emerge in the Sāṅkhya creation sequence.

_____ → _____ → _____ → _____ → _____

2. (3 marks) Complete the quality table.

Element	Number of qualities
<i>ākāśa</i>	_____
<i>vāyu</i>	_____

Element	Number of qualities
<i>tejas</i>	_____
<i>ap</i>	_____
<i>pṛthvī</i>	_____

3. (2 marks) Match each element to its *tanmātra*. Draw lines.

ākāśa	gandha (smell)
vāyu	śabda (sound)
tejas	rasa (taste)
ap	sparśa (touch)
pṛthvī	rūpa (form)

4. (1 mark) Name one object in this room dominated by *ākāśa*.

5. (2 marks, short answer) Why does it make sense that *pṛthvī* has all five qualities?

Part B — answer key

Q	Answer	Mark scheme
1	ākāśa → vāyu → tejas → ap → pṛthvī	2 if all 5 in correct order; 1 if 3–4 correct; 0 otherwise.
2	1, 2, 3, 4, 5	3 if all correct; 2 if 3–4 correct; 1 if 1–2 correct.
3	ākāśa ↔ śabda, vāyu ↔ sparśa, tejas ↔ rūpa, ap ↔ rasa, pṛthvī ↔ gandha	2 if all 5; 1 if 3–4.
4	The empty space between two students; the air-gap above their head; the gap in a doorway; any answer that names a <i>void</i> not a thing.	1 if it names empty space rather than an object.
5	Earth is the last element to emerge, so it inherits all qualities from the elements before it (sound, touch, form, taste) and adds its own (smell). You can perceive a piece of soil with every sense.	2 if the answer mentions both <i>inherits previous</i> AND <i>adds smell</i> . 1 if either part is present.

Total Part B: 10 marks. Not added to summative.

What to do with the results

- A class average below 6/10 on Part B = significant gaps. Reteach the quality-count rule and tanmātra map on Day 6 (push the cross-disciplinary content to Day 7).
- Individual students below 3/10 on Part B = pull aside on Day 7 for a 10-min one-on-one before group activity starts.

Module 2 (Middle) — Summative Quiz

Time: 20 min · **Marks:** 25 · **Closed book**

Instructions: Write your name and class. Answer all questions. Section D has one mark for clear handwriting and sentence structure. No calculators needed.

Name: _____ **Class:** _____

Section A — Multiple choice ($5 \times 1 = 5$ marks)

1. Which of the following elements emerged FIRST according to the Sāṅkhya sequence?

- a) *pr̥thvī*
- b) *ap*
- c) *ākāśa*
- d) *tejas*

2. The *tanmātra* (subtle quality) of the fire element is:

- a) *śabda*
- b) *sparsā*
- c) *rūpa*
- d) *gandha*

3. Which element does the tongue primarily perceive?

- a) *ākāśa*
- b) *vāyu*
- c) *ap*
- d) *pr̥thvī*

4. Which of these does the *pañcabhūta* framework include that the modern periodic table does NOT?

- a) Hydrogen
- b) Water
- c) Empty space

d) Fire

5. True/false (circle one): The five elements are arranged from greatest number of qualities (5) to fewest (1) in the order *prthvī*, *ap*, *tejas*, *vāyu*, *ākāśa*. T / F

Section B — Short answer (4 × 2 = 8 marks)

6. Write the five elements in the order they emerge, with their English meanings.

7. Why does the tradition say *ākāśa* has only one quality? What is that quality?

8. Give one example each of an object dominated by *tejas* and an object dominated by *vāyu*.

9. Name one tanmātra and the sense organ that perceives it.

Section C — Diagram + working (1 × 5 = 5 marks)

10. Draw the *pañcabhūta* creation sequence as nested layers (or stacked boxes — your choice). Label each layer with (a) the Sanskrit name, (b) the English meaning, (c) the number of qualities.

[draw in the space below]

Section D — Open response (1 × 6 = 6 marks; 1 of which is for clarity)

11. Pick ONE element. In 4–5 sentences, describe: - one place near your home where this element is in *excess* or *depleted* - what observable consequence you’ve seen - one specific action that you (or your community) could take to restore balance

Answer key (teacher copy)

Q	Answer	Mark scheme
1	c) ākāśa	1
2	c) rūpa	1
3	c) ap	1
4	c) Empty space	1
5	F (the order is reversed — emergence is ākāśa → pṛthvī, but pṛthvī has the most qualities)	1; accept either as correct if reasoning shown
6	ākāśa (space) → vāyu (air) → tejas (fire) → ap (water) → pṛthvī (earth)	2 if all in correct order with English; 1 if order correct but English missing or vice versa.
7	Because it is the first to emerge — it has no element	2 if both <i>why</i> and <i>what</i> present. 1 if only one.

Q	Answer	Mark scheme
	before it to inherit qualities from. The one quality is <i>śabda</i> (sound).	
8	tejas: candle/flame/sunlight/tube light. vāyu: balloon/breeze/breath/balloon -pump.	2 (1 each)
9	Any of the five pairs: ākāśa ↔ ear, vāyu ↔ skin, tejas ↔ eye, ap ↔ tongue, pṛthvī ↔ nose.	2 (1 for tanmātra named correctly, 1 for matching organ)
10	A nested or stacked diagram with all five elements correctly labelled in order and quality-counts (1, 2, 3, 4, 5).	5: 2 for correct order, 2 for quality counts, 1 for clear labelling.
11	A specific local example, observable consequence, and concrete action. Imbalance must match the named element.	6: 2 for specific local example; 2 for observable consequence; 1 for plausible action; 1 for clarity of writing.

Total: 25 marks.

Grading bands (for project combined): - 32–40: Exceeds - 24–31: Meets - 16–23: Approaches - 0–15: Beginning

Homework

Homework — Module 2 Panchabhūta (Middle)

Homework tasks per day. None should take more than 25 minutes for a Middle-band student. Skip a day if it conflicts with another subject's deadline.

Day 1 → Day 2

Task: Bring **three interesting objects from home** to class on Day 4. They should be small enough to fit on your desk.

Bonus: Which element does each object belong to?

Time: 5 min (deciding) + carrying the objects.

Day 2 → Day 3

Task: Find ONE thing in your home that perfectly demonstrates the “all five qualities” of *pṛthvī*. It should be something you can hear, touch, see, taste (or imagine the taste), and smell. Write 2 sentences naming the object and what each quality feels like.

Examples your classmates might pick: a sliced lemon, a piece of jaggery, a freshly cut carrot, a hand-made clay pot.

Time: 10 min.

Day 3 → Day 4

Task: For each of your three home-objects, figure out the **dominant element** and the **dominant tanmātra**. Write one line per object.

Format: Object → dominant element → dominant tanmātra

Example: Lemon → pṛthvī → gandha (smell) is strongest, but rasa (taste) is close behind.

Time: 10 min.

Day 4 → Day 5

No new homework. Use the time to finalise your project pair / individual decision and to start thinking about which element you want to focus on.

Day 5 → Day 6

Task: Submit your **project plan paragraph** in writing.

Format: one paragraph, 5–8 sentences, stating: which element, which format (poster / demo / story / video / infographic / song), and what audience will see/hear/feel.

Time: 15 min.

Day 6 → Day 7

Task: Open your school science textbook to the chapter on “Matter in Our Surroundings” (Class 9, Chapter 1) — or whatever your closest equivalent is. Read one section. Write a 1-paragraph note: *what would the pañcabhūta framework get right about this topic, and what would it miss?*

Time: 20 min.

Day 7 → Day 8

Task: Bring ALL project materials to tomorrow’s class. The build session starts immediately. Anything you forget cannot be retrieved during class time.

Bonus: Pick one of the five elements (not your project element) and identify one “imbalance” example near your home — a polluted drain (*ap*), a dusty lot (*prthvī*), a noisy intersection (*ākāśa+śabda*), etc. Be ready to share in Day 7’s group session.

Time: 10 min.

Day 8 → Day 9

Task: Finish at home what you couldn’t finish in class today. Tomorrow is for *refining*, not for *starting*.

Time: variable — but cap at 30 min, then stop and rest. Done-is-better-than-perfect.

Day 9 → Day 10

Task: Final touches. Make sure you have: - [] The artefact (poster / video file / printed story / etc.) - [] Your script or notes - [] A working version of your 3-minute presentation in your head

Time: 15 min.

After Day 10

No homework. Module is complete. Take a breath. (If your next module begins immediately, check that module’s README for prerequisite reading.)

Writing Prompts

Writing Prompts — Module 2 Panchabhūta (Middle)

Eight prompts in mixed modes. Assign as in-class quick-writes (10 min), homework (one paragraph), or as journal entries woven across the module. Each prompt names the lesson day where it fits naturally.

Reflection prompts

- 1. (Day 1, 5 min)** Pick the element that surprised you most today. In 3–5 sentences, describe a moment from your morning when this element was around you and you didn't notice it. Now that you've noticed, what changes?
- 2. (Day 7, 10 min)** Sit somewhere quiet for two minutes (you can do this in class or at home before writing). Then list the five elements and rank them by which one is *most present* and which is *least present* where you sat. Write 4 sentences explaining your top and bottom ranking.

Explanatory prompts (in your own words)

- 3. (Day 2 or 3, homework, ~150 words)** Explain to a younger sibling (Grade 4) why earth has all five qualities but space has only one. Don't use the word "tanmātra" — find your own way to say it.
- 4. (Day 6, in-class, 10 min)** Imagine you are a chemistry teacher being asked about *pañcabhūta* by a curious Grade 10 student. In 5 sentences, answer the question: "*Is the pañcabhūta framework outdated?*" Be fair to both frameworks.

Argumentative prompt

- 5. (Day 6 or Day 7, homework, 1 paragraph)** A modern science textbook says fire is "not an element" — it is hot gas with light from incandescent particles. The *pañcabhūta* tradition says *tejas* IS an element. Pick a side and defend it in 5–7 sentences. Acknowledge the other side's strongest point before disagreeing.

Creative prompt

- 6. (Day 5 or Day 8, can be project — 1 page)** Write a short scene (50–150 words) in which the five elements have a conversation about who is the most important. Each element gets at least one line. End with one element pointing out something the others have missed.

Cross-disciplinary prompt

- 7. (Day 6 or homework, ~120 words)** Pick a topic from another subject — geography, biology, music, sport, anything. Show how the *pañcabhūta* lens would describe one feature of that topic that the standard subject vocabulary misses. (Example: in music, *ākāśa* is the silence between two notes; in football, *vāyu* is the wind that bends a free kick.)

Stretch prompt

- 8. (Optional, advanced students, Day 10 onwards, ~250 words)** The Greek philosopher Empedocles (5th century BCE) proposed FOUR elements: earth, water, fire, air. There is no equivalent of *ākāśa* in his scheme. Aristotle later added a "fifth essence" — but it was the substance of the heavens, not of space-in-general. In a single page, argue for OR against the claim: "*Including ākāśa makes the Indian framework more powerful than the Greek one.*" Use one specific example to support your argument.

Prompt-to-band notes

- All prompts are usable at Middle band as written.
- For **Primary** band, prompts 1, 2, 3, and 6 can be adapted (shorter, drawing-friendly).
- For **Senior** band, prompts 4, 5, 7, and 8 are essay-quality without modification; add a citation requirement.

How to give feedback

- Short prompts (1, 2): respond with one sentence of acknowledgement, not corrections. The point is observation.
- Explanatory (3, 4): respond with “the part that landed” and “the part still unclear.” Don’t correct grammar unless severely impeding meaning.
- Argumentative (5, 8): use the rubric — does the writing acknowledge the other side, give a specific example, and reach a clear position?
- Creative (6): no grading. Read aloud the best ones if you have permission.

Activity 01 — Element Sort

Module: Panchabhūta · **Band:** Middle · **Day:** 4 · **Time:** 25 min (within Day 4’s 45 min)

Type

Hands-on classification, paired

Goal

Each pair sorts 5 real objects into the five-element framework and documents which other elements are also present in each.

Materials per pair

- 3 objects from each student’s home (5–6 objects per pair after duplicates)
- 2 mystery objects supplied by the teacher per pair (e.g. a stretched balloon, a glass of cloudy water, a battery, a piece of camphor, a chunk of beeswax, a small mirror)
- 1 printed worksheet (next page)
- 1 pen each

Set-up (teacher, before class)

- Lay out 5 tray-areas around the room, each clearly labelled with one element name (Sanskrit + English).
- Distribute 2 mystery objects to each pair as they enter.

Instructions (student-facing — printable handout)

Element Sort — work in pairs.

1. Lay out your 5 objects on your desk.
2. For each object, decide which element is **dominant** (the strongest match). Place the object on the matching element tray.
3. Fill in the worksheet table below for all 5 objects.
4. **Disagree with your partner?** Argue it out before you fill in the row. If you can't agree, write both answers and circle the one your pair finally votes for.
5. **Gallery walk (last 5 min):** visit two other pairs' trays. If you disagree with where they placed an object, write a sticky note explaining why (kindly).

Worksheet (one per pair)

#	Object	Dominant element	Other elements present	Strongest <i>tanmātra</i>	Disagreement notes
1					
2					
3					
4					
5					

Discussion prompts (last 5 min, whole class)

- Which object was the hardest to classify? Why?
- Was there an object that two pairs put on different trays? What was the strongest argument for each side?
- Did any object seem to fit on *no* tray? What does that tell us?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Both partners are pointing at the object, talking it through, sometimes disagreeing then resolving	"What about the smell?" / "Yes but the form is more obvious" / "Let's say dominant is <i>prthvī</i> but <i>gandha</i> is the strongest <i>tanmātra</i> "
Needs intervention	One partner doing all the writing while the other sits silently; rows filled in with one-word answers and no thought	Silence; "I don't know, write whatever"

Safety / sensitivity

- No fire / no flame in this activity (despite *tejas* — students can find heat sources without lighting anything).

- No food sharing for tasting; if a student wants to demonstrate *rasa*, they taste their own food, not a classmate's.
- Allergy check before allowing mystery objects like peanut shells, citrus peel, etc.

Differentiation

- **Tier up:** After sorting, the pair picks their single “most mixed” object and writes a paragraph allocating percentages to each element (e.g. 50% *pṛthvī*, 30% *ap*, 15% *vāyu*, 5% *tejas*, 0% *ākāśa*). Defend the percentages.
- **Tier down:** Sort first into THREE categories (*pṛthvī*, *ap*, *vāyu*) and come back to add *tejas* and *ākāśa* once the first three are confident.

Clean-up

- Each pair takes their own objects home.
- Teacher resets the 5 trays for the next class.
- Worksheet collected (stored in the project folder for Day 8 reference).

Activity 02 — Sense Station Rotation

Module: Panchabhūta · **Band:** Middle · **Day:** 3 · **Time:** 15 min (within Day 3's 45 min)

Type

Hands-on sensory perception, paired

Goal

Students experience each *tanmātra* directly and notice when more than one perception is happening simultaneously.

Materials per group

5 stations (one per element), each set up with:

- **Station *ākāśa* (sound)** — a small bell or two metal spoons. A printed label: “*Perceive sound. Use your ears.*”
- **Station *vāyu* (touch)** — a piece of cotton wool, a small ice pack (or cold water in a sealed bag), a piece of sandpaper. Label: “*Perceive touch. Use your skin.*”
- **Station *tejas* (form / colour)** — a small mirror, a coloured plastic sheet, an unlit torch turned on. Label: “*Perceive form. Use your eyes.*”
- **Station *ap* (taste)** — a tiny pinch of salt, a tiny pinch of sugar (or jaggery), on individual paper plates. (Or a jar of mint leaves to chew if no food allowed.) Label: “*Perceive taste. Use your tongue.*”
- **Station *pṛthvī* (smell)** — a sprig of mint, a slice of orange peel, a small jar of soil after wetting. Label: “*Perceive smell. Use your nose.*”

Set-up (teacher, before class)

- Arrange 5 stations around the room.
- Each pair is assigned a starting station (rotate clockwise every 2 min).
- 5 stations × 2 min each = 10 min for rotation + 5 min discussion.

Instructions (student-facing — printable handout)

Sense Station Rotation — work in pairs.

At each station, spend 2 minutes:

1. Use the sense organ named on the label to perceive the sample.
2. On your worksheet, write one sentence describing **what you perceived**.
3. Then ask yourself: **what OTHER sense could perceive something here at the same time?** Write that too.

Rotate when the teacher rings the bell.

Worksheet (one per pair)

Station	Tanmātra	What I perceived	Other element / tanmātra I noticed
ākāśa	<i>śabda</i> (sound)		
vāyu	<i>sparsa</i> (touch)		
tejas	<i>rūpa</i> (form)		
ap	<i>rasa</i> (taste)		
pṛthvī	<i>gandha</i> (smell)		

Discussion prompts (last 3 min)

- At which station did you notice *more than one* element at once?
- Example: at the bell — did you HEAR the sound (*ākāśa*) and also FEEL the vibration in your fingers if you touched the bell (*vāyu*)?
- What does this tell us about how the elements actually appear in the world?

Safety / sensitivity

- **Diabetic check** before using sugar / jaggery — substitute with salt or a non-edible smell-only sample.
- **Allergy check** for peel, mint, soil.
- No sharing of food samples between students.
- Cold pack: don't let students hold it for more than 30 seconds.

Differentiation

- **Tier up:** At each station, name ONE everyday object that demonstrates the same tanmātra better. (E.g. ākāśa station — “I hear better with a singing bowl than this bell.”)
- **Tier down:** Just write what you perceived. Skip the “other element” column.

Clean-up

- Teacher collects all materials; students do not eat from leftover samples.

Activity 03 — Element Walk (Field Observation)

Module: Panchabhūta · **Band:** Middle · **Day:** 7 (debriefed in class; observation done at home overnight)

Type

Independent field observation, 20 min, completed at home or on the school grounds

Goal

Each student finds ONE example of an element imbalance in their neighbourhood and documents it.

Materials per student

- Notebook + pen, or a smartphone for photos
- The element-walk worksheet (below)

Instructions (student-facing — printable handout)

Element Walk

Spend 20 minutes walking around your neighbourhood (or the school grounds). Look for ONE place where one of the five elements is clearly out of balance — too much, or too little.

Examples of what to look for:

Element	Excess	Depletion
<i>pṛthvī</i> (earth)	Pile of plastic waste; bald soil from no plants	Eroded soil; bare slope
<i>ap</i> (water)	Stagnant water pool (mosquitoes); flooded drain	Dry tap; cracked dry earth
<i>tejas</i> (fire/heat)	Heat-island parking lot at noon; smoke from burning garbage	Cold dark corner where nothing grows
<i>vāyu</i> (air)	Vehicle exhaust at a	Stagnant indoor air; no breeze in a

Element	Excess	Depletion
	signal; dust storm	closed alley
<i>ākāśa</i> (space/sound)	Constant honking; loudspeakers	An eerily silent abandoned spot

Document on the worksheet:

Worksheet (one per student)

Date: _____ Time of observation: _____

Location (one line): _____

Which element is out of balance? _____

Is it in EXCESS or DEPLETION? (circle one) EXCESS / DEPLETION

What did you observe? (3 sentences)

What is one observable consequence?

What is ONE small action you (or your community) could take?

(Optional) Sketch or describe what you saw:

Discussion prompts (Day 7 in class)

- Whose observation was the most specific (named a real street, a real time)?
- Did any two students choose the same location? How did their observations differ?
- Which “small actions” feel most doable? Which feel hardest?

Safety / sensitivity

- **No solo walks for younger students (Grade 6).** Pair with a sibling, parent, or do the walk on school grounds during break.
- **Avoid actually inhaling smoke, drinking unsafe water, etc.** Observation is the goal — not exposure.

- **Be aware** of sensitivity around naming specific neighbourhoods as polluted (some students live there). Frame the observation as “everyone’s neighbourhood has imbalances, including mine.”

Differentiation

- **Tier up:** Find TWO related imbalances (e.g. polluted water leads to depleted plant life). Document the connection.
- **Tier down:** Just describe what you saw. The “one action” can be one of the suggestions from the in-class examples.

Clean-up

- Bring the worksheet to Day 7 class.
- Photos optional — only if devices are allowed in school per school policy.

Project Brief

Project Brief — Module 2 Panchabhūta (Middle)

Work sessions: Day 8 + Day 9 (in class) · **Presentation + summative:** Day 10 · **Total marks:** 40

The project

Each pair (or individual, with teacher approval) chooses **one** of the five elements and produces a small project that shows:

1. **What the *pañcabhūta* tradition says** about this element (its tanmātra, its position in the creation sequence, the quality-count rule, what it “does”). Must cite at least one source — a verse from the RAG cache, or a reference from `sources.md`.
2. **One modern-science parallel** — a concept from chemistry, biology, physics, or ecology that overlaps with this element. Explain where the parallel works and where it breaks down.
3. **One local imbalance** — a real example in your neighbourhood where this element is in excess or depletion, with one doable action.

Why this matters

You’re learning to hold two intellectual frameworks side-by-side without forcing one to win. This is a core skill — not just for *pañcabhūta*, but for any time you encounter a way of seeing the world that’s different from your own.

What success looks like

A strong project will: - Demonstrate the module’s central concepts accurately (tanmātra, creation sequence, quality-count rule). - Connect the Indian tradition to a modern scientific concept fairly

— without saying “ancient Indians knew this already.” - Show your own thinking, not just summary. - Be clearly communicated in your chosen medium.

Format options

Format	Best for elements	Notes
Poster (A2 or larger)	Any	Lots of visual space; works for nervous presenters
Live demonstration (2 min hands-on)	<i>tejas</i> (fire), <i>vāyu</i> (air), <i>ap</i> (water)	Needs safety check
Short written story (1 page)	<i>ākāśa</i> , any	Read aloud or hand out
90-second video	Any	Pre-made and played in class
Infographic	<i>pṛthvī</i> (data-heavy), <i>ap</i>	Combine drawing + facts
Original song or poem	<i>vāyu</i> , <i>ākāśa</i>	90 sec performed live

Other formats with teacher approval. Whatever the format, the 3-minute time limit on Day 10 applies.

Working in pairs (or individually)

- **Pair work is encouraged**, especially for video/song/demo formats.
- **Individual is allowed**, especially if the topic is *ākāśa* (often suits a solo reflective piece).
- Groups of 3 are acceptable but each member must have a distinct sub-role (visuals / script / presenter, or research / draft / final).

Schedule

Day	What you do	Submit at end of class
5	Choose element + format + partner	Sticky note on the board
6	(Homework) Submit project plan paragraph	Project plan paragraph
8	Plan + start first artefact	Draft of first artefact (photo or scan accepted)
9	Refine + rehearse	Draft of final artefact
10	Present (3 min) + Q&A (1 min)	Final artefact + reflection slip

Citation rules

- At minimum **one** citation must appear in your project — visible on the poster, named in the video, or printed in the story.
- The citation must be from a source you can point to: a verse from one of the day-NN.json cache files in `.rag-cache/`, OR a reference from `sources.md`.
- If you invent a citation or pull one off the internet without verification, you'll score 0 on both **Use of sources** and **Originality**.

Rubric

See `rubric.md`. Read it before Day 8.

Self-reflection (submitted on Day 10 with project)

In 4–6 sentences, answer:

1. What did you set out to learn or make, and what did you actually end up with?
2. What is the most surprising thing you discovered while making this?
3. If you had three more days, what would you do differently?

Exemplars

A “Meets” exemplar will be added to `exports/pdf/middle-module-02-panchabhuta/exemplar-strong.pdf` after the first cohort presents.

Rubric

Project Rubric — Module 2 Panchabhūta (Middle)

Project marks: 30 (out of 40 total when combined with presentation 10).

Criteria

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
Conceptual accuracy	Names the element AND its tanmātra AND its position in the creation sequence correctly. Connects to the quality-	Names the element, tanmātra, and one accurate fact about its place in the framework. Minor	Names the element correctly but mixes up tanmātra OR creation order.	Names the element only; gets the tanmātra or creation order wrong.	× 2

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
	count rule.	imprecision OK.			
Use of sources	Cites at least one verse from the RAG cache verbatim, with reference (e.g. “SB 3.26.36”); cites it correctly. Bonus: weaves a paraphrase in addition to the direct cite.	Cites a source by reference; the source matches a real entry in the cache or sources.md.	Mentions a source vaguely (“an old Sanskrit text says ...”) without a specific reference.	No source cited.	× 2
Modern-science parallel	Names a specific modern-science concept and explains the parallel AND where it falls short.	Names a modern-science concept and explains the parallel.	Names a modern-science concept but doesn’t explain the parallel.	No parallel attempted.	× 1
Local-imbalance connection	Names a specific local example with date/place/observation + names a doable action.	Names a local example and an observation .	Names a generic example (“pollution is bad”).	No local connection.	× 1
Originality of expression	The project shows the pair’s own thinking — a turn of phrase, a fresh comparison, a surprising example.	Clearly the students’ own work; not lifted from any single source.	Largely paraphrased from a single source; little independent thinking visible.	Clearly copied from a single source.	× 1
Communication /	Visuals, layout, or	Communicates clearly.	Message comes through despite	Hard to follow; multiple craft	× 1

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
craft	narrative are clean, deliberate, and serve the message. Spelling and IAST are correct.	One or two craft issues (a typo, a misaligned heading).	some craft issues.	issues impede understanding.	

Score calculation

Project score = (sum of criterion scores × weights) × 0.9375

Maximum raw score: 32. Multiply by 30/32 = 0.9375 to get out of 30.

Presentation marks (10, on Day 10)

Criterion	4	3	2	1	Weight
Within time (3 min)	Within 2:45–3:00	Within 2:30–3:15	Within 2:15–3:30	Outside that range	× 1
Both partners contribute (or solo presents whole)	Equal contribution; smooth handoffs	Both partners speak but unequally	One partner dominates	One partner silent	× 1
Clarity and audibility	Confident, audible to the back of the room, clear pace	Audible, mostly clear	Audible only in front rows; some mumbling	Mostly inaudible	× 0.5

Maximum: 10.

Self-reflection mark (folded into project score)

The 3 reflection questions are not separately scored — but if the reflection is missing or perfunctory, the **Originality** mark drops by 1.

Common decision points

- **Group of 3 vs pair:** Acceptable, but the third member must have a clearly distinct sub-role (e.g. visuals + script + presenter). Marks are equal across the group unless one member visibly didn't contribute.
- **Late submission:** -2 marks per school day, capped at -6.
- **Originality dispute:** If a project closely matches an online source not cited, the project goes to 0 on **Use of sources** AND on **Originality**. A 1-day extension is offered for a redo.

Marking process

- Mark in real-time during presentation (5 minutes per pair: 3 present + 1 Q&A + 1 marking).
- Write 1-sentence written feedback after each presentation finishes.
- Return marked rubrics within 2 school days.